

SPHERE PACKING PROBLEM

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ABSTRACT. Delsarte introduced linear programming bounds to bound the size of error-correcting codes. Cohn and Elkies adapted a similar approach to provide upper bounds for sphere packings. They conjectured that these bounds are sharp enough to prove the optimal packing arrangements in dimensions 8 and 24, which was later resolved by Maryna Viazovska and collaborators. We survey Delsarte's program, the Cohn–Elkies theorem, and explain the main ideas behind Viazovska's construction of the magic function proving the optimal sphere packing bound in dimension 8.

1. INTRODUCTION

1.1. Packing Problem. Let $B_d(x, r)$ denote the Euclidean ball centered at $x \in \mathbb{R}^d$ with radius r . A sphere packing is an arrangement of congruent balls with disjoint interiors:

$$\mathcal{P} = \bigcup_{x \in X} B_d(x, r)$$

The finite density of $\mathcal{P}$ is defined by

$$\Delta_{\mathcal{P}}(R) = \frac{\text{vol}(B_d(0, R) \cap \mathcal{P})}{\text{vol}(B_d(0, R))}$$

To measure the portion of $\mathbb{R}^d$ occupied by $\mathcal{P}$, the upper density is defined by

$$\Delta_{\mathcal{P}} = \limsup_{R \rightarrow \infty} \Delta_{\mathcal{P}}(R) = \limsup_{R \rightarrow \infty} \frac{\text{vol}(B_d(0, R) \cap \mathcal{P})}{\text{vol}(B_d(0, R))}$$

This quantity is independent of the radius of the balls in the packing.

Question 1.1. What is the optimal density among all possible packings?

$$\Delta_{\mathbb{R}^d} = \sup_{\mathcal{P}} \Delta_{\mathcal{P}}$$

The optimal density is simply 1 in one dimension. In two dimensions, the optimal packing is the hexagonal packing. In three dimensions, the optimal density is attained by the face-centered cubic packing and by related stackings of hexagonal layers.

A greedy argument gives the pessimistic lower bound 2^{-d} for the optimal density. Indeed, consider a packing $\mathcal{P}$ of unit balls that has no room for another unit ball. Let

$$X_R := \{x \in X : \|x\| \leq R + 2\}$$

By maximality, enlarging each ball centered at a point of X_R by a factor of 2 covers $B_d(0, R)$. Hence

$$\text{vol}(B_d(0, R)) \leq \text{vol} \left(\bigcup_{x \in X_R} B_d(x, 2) \right) \leq \sum_{x \in X_R} \text{vol}(B_d(x, 2)) = 2^d \sum_{x \in X_R} \text{vol}(B_d(x, 1))$$

Thus these unit balls occupy at least a 2^{-d} fraction. Taking $R \rightarrow \infty$, we obtain

$$\Delta_{\mathbb{R}^d} \geq 2^{-d}$$

1.2. Error-Correcting Codes. A closely related question comes from coding theory: what is the maximal size of a *code* $C \subset \mathbb{F}_2^n$ such that distinct codewords are far apart? Here distance is measured by the Hamming distance $\rho(x, y)$. For example, every pair of codewords in

$$C = \{01011, 11100, 00110\} \subset \mathbb{F}_2^5$$

differs by at least 3 digits. Suppose you send 01011 to your friend, who wrongly receives 01111. Since 01111 is closest to 01011 among the codewords in C , your friend can correct it back to 01011. This is the idea of *error-correcting codes*: if the codewords in C are sufficiently far apart, a small distortion can be corrected by choosing the closest codeword.

More generally, suppose we want to send a codeword of length n from a subset $C \subset \mathbb{F}_2^n$, with possible distortion in roughly $d/2$ digits. What is the maximal size of C so that this error-correcting procedure always works?

Question 1.2 (Delsarte's Extremal Problem). For $n, d \in \mathbb{N}$, what is the value of

$$A(n, d) = \max\{|C| : C \subset \mathbb{F}_2^n, \forall x \neq y \in C, \rho(x, y) \geq d\}?$$

For any such subset C , the error-correcting procedure works as long as the number of distorted digits is at most

$$\left\lfloor \frac{d-1}{2} \right\rfloor$$

Sphere packing in Euclidean space is a continuous analogue of Question 1.2: what is the maximal size of a set of centers $C \subset B_R(0)$ such that

$$\forall x \neq y \in C, \quad \|x - y\|_2 \geq 2?$$

Both of these packing problems can be formulated generally as follows.

Question 1.3. Let (M, ρ) be a metric space. We say $C \subset M$ is an r_0 -code if

$$\forall x \neq y \in C, \quad \rho(x, y) \geq r_0$$

What is the maximal size of an r_0 -code?

We say a function $f : \text{Im}(\rho) \rightarrow \mathbb{R}$ is copositive if, for every finite $X \subset M$,

$$\sum_{x, y \in X} f(\rho(x, y)) \geq 0$$

If we can find copositive functions that are sufficiently negative for all distances $\geq r_0$, then we can bound the maximal size of an r_0 -code.

Theorem 1.4. Suppose there exists a copositive function $f : \text{Im}(\rho) \rightarrow \mathbb{R}$ such that

- (1) $f(0) = 1$,
- (2) $f(t) \leq \frac{-1}{N-1}$ for $t \geq r_0$.

Then for any finite r_0 -code $X \subset M$, we have $|X| \leq N$.

Proof. Let X be a finite r_0 -code. By copositivity of f ,

$$\sum_{x,y \in X} f(\rho(x,y)) \geq 0$$

On the other hand, breaking the same sum into diagonal and off-diagonal terms gives

$$\sum_{x,y \in X} f(\rho(x,y)) = \sum_{x \in X} f(\rho(x,x)) + \sum_{\substack{x,y \in X \\ x \neq y}} f(\rho(x,y)) \leq |X|f(0) - \frac{|X|(|X| - 1)}{N - 1}$$

Combining the previous two inequalities yields $|X| \leq N$. $\square$

This may seem miraculous at first glance. In the remainder of this section, we motivate the theorem from Delsarte's Fourier-analytic approach to error-correcting codes.

1.3. Boolean Fourier Analysis. The Boolean cube $\mathbb{F}_2^n = \{0,1\}^n$ is a finite abelian group with inner product

$$\langle f, g \rangle = \frac{1}{2^n} \sum_{x \in \mathbb{F}_2^n} f(x) \overline{g(x)}$$

A set of orthogonal characters is

$$\chi_s(x) = (-1)^{s \cdot x}, \quad s \in \mathbb{F}_2^n$$

which is analogous to $e^{2\pi i \xi \cdot x}$ in $\mathbb{R}^n$.

1.4. Distance Distribution. For a code $C \subset \mathbb{F}_2^n$, we can look at the distribution of pairwise distances:

$$A_k := \frac{1}{|C|} \left| \{(x,y) \in C^2 : \rho(x,y) = k\} \right|, \quad 0 \leq k \leq n$$

We collect some basic properties of A_k . By definition,

$$(1) \quad A_0 = 1, \quad A_k \geq 0, \quad \sum_{k=0}^n A_k = |C|$$

If C is an r_0 -code, where $0 \leq r_0 \leq n$, then

$$(2) \quad A_1 = A_2 = \cdots = A_{r_0-1} = 0$$

Delsarte proposes that instead of looking at specific codes, we should consider their distance distributions $A = (A_0, A_1, \dots, A_n)$. This is a useful relaxation, at the cost that many vectors $A \in \mathbb{R}^{n+1}$ are not actually attained as the distance distribution of any code C .

The character $\chi_s(x)$ currently depends on s . If we group the characters by their weight $|s| = j$ and fix any $z \in \mathbb{F}_2^n$ with $|z| = i$, define

$$K_j(i) = \sum_{\substack{s \in \mathbb{F}_2^n \\ |s|=j}} \chi_s(z) = \sum_{\substack{s \in \mathbb{F}_2^n \\ |s|=j}} (-1)^{s \cdot z}$$

By symmetry of $\mathbb{F}_2^n$, the right-hand side is independent of the specific choice of z and depends only on $|z| = i$.

Proposition 1.5 (Krawtchouk polynomials).

$$K_j(i) = \sum_{k=0}^j (-1)^k \binom{i}{k} \binom{n-i}{j-k}$$

Proof. If z has i ones and $n-i$ zeros, and s has j ones, suppose s overlaps with z in exactly k positions. Then we choose k overlapping positions from the i ones of z , and choose the remaining $j-k$ ones from the $n-i$ zero positions of z . The sign is $(-1)^k$. $\square$

We now use the positivity

$$\sum_{x,y \in C} \chi_s(x-y) = \sum_{x,y \in C} \chi_s(x) \overline{\chi_s(y)} = \left| \sum_{x \in C} \chi_s(x) \right|^2 \geq 0$$

Summing this over all frequencies s with $|s| = j$, we obtain

$$\sum_{\substack{s \in \mathbb{F}_2^n \\ |s|=j}} \sum_{x,y \in C} \chi_s(x-y) \geq 0$$

Swapping the order of summation gives

$$\sum_{x,y \in C} \sum_{\substack{s \in \mathbb{F}_2^n \\ |s|=j}} \chi_s(x-y) \geq 0$$

As we saw from the Krawtchouk polynomial, the inner sum depends only on the Hamming distance $\rho(x,y)$, and is equal to $K_j(\rho(x,y))$. Therefore

$$\sum_{x,y \in C} K_j(\rho(x,y)) \geq 0$$

Now group the ordered pairs (x,y) according to their distance. By definition of the distance distribution A_i , there are $|C|A_i$ ordered pairs at distance i . Hence

$$\sum_{x,y \in C} K_j(\rho(x,y)) = |C| \sum_{i=0}^n A_i K_j(i)$$

Since $|C| > 0$, we obtain

$$\boxed{\sum_{i=0}^n A_i K_j(i) \geq 0, \quad j = 0, \dots, n.}$$

We conclude that the Krawtchouk transform of the distance distribution is non-negative.

1.5. Delsarte's Linear Program. We now assemble the constraints. A code $C \subset \mathbb{F}_2^n$ with minimum distance at least d gives a vector

$$A = (A_0, \dots, A_n)$$

satisfying

$$\begin{aligned} A_0 &= 1, \\ A_i &= 0, \quad 1 \leq i < d, \\ A_i &\geq 0, \end{aligned}$$

and

$$\sum_{i=0}^n A_i K_j(i) \geq 0, \quad j = 0, \dots, n$$

Moreover,

$$\sum_{i=0}^n A_i = |C|$$

Thus we obtain the following linear program:

$$\begin{array}{ll} \text{maximize} & \sum_{i=0}^n A_i \\ \text{subject to} & A_0 = 1, \\ & A_i = 0, \quad 1 \leq i < d, \\ & A_i \geq 0, \\ & \sum_{i=0}^n A_i K_j(i) \geq 0, \quad j = 0, \dots, n. \end{array}$$

Every genuine code gives a feasible point of this linear program. Therefore

$$A(n, d) \leq \text{optimal value of the Delsarte linear program}$$

1.6. Dual Program. The dual point of view is closer to the later Cohn–Elkies method. Instead of maximizing over possible distance distributions, we look for a polynomial that certifies an upper bound for every possible code. Let

$$p(i) = \sum_{j=0}^n c_j K_j(i),$$

where

$$c_j \geq 0, \quad c_0 = 1$$

Assume moreover that

$$p(i) \leq 0, \quad d \leq i \leq n$$

Then for any code C of size $M = |C|$, we have

$$\sum_i A_i p(i) = \sum_j c_j \sum_i A_i K_j(i)$$

By the Delsarte inequalities and the nonnegativity of the coefficients c_j ,

$$\sum_i A_i p(i) \geq c_0 \sum_i A_i K_0(i)$$

Since $K_0(i) = 1$, this becomes

$$\sum_i A_i K_0(i) = \sum_i A_i = M$$

Thus

$$\sum_i A_i p(i) \geq M$$

On the other hand, using $A_0 = 1$, $A_i = 0$ for $1 \leq i < d$, and $p(i) \leq 0$ for $i \geq d$, we get

$$\sum_i A_i p(i) = p(0) + \sum_{i=d}^n A_i p(i) \leq p(0)$$

Therefore

$$M \leq p(0)$$

Since

$$K_j(0) = \binom{n}{j},$$

we have

$$p(0) = \sum_{j=0}^n c_j \binom{n}{j}$$

Thus the dual linear program is

$\begin{aligned} &\text{minimize} && \sum_{j=0}^n c_j \binom{n}{j} \\ &\text{subject to} && \sum_{j=0}^n c_j K_j(i) \leq 0, \quad d \leq i \leq n, \\ &&& c_j \geq 0, \\ &&& c_0 = 1. \end{aligned}$
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Delsarte's dual problem asks us to find a test polynomial with nonnegative Krawtchouk, or Fourier, coefficients and negative values at all allowed nonzero distances. Cohn–Elkies adapted this to sphere packing in $\mathbb{R}^n$ by asking for a Schwartz test function with nonnegative Fourier transform and negative values at all allowed nonzero Euclidean distances.

2. LATTICE AND PACKING

A lattice in $\mathbb{R}^d$ is the $\mathbb{Z}$ -linear span of a basis $\{v_1, v_2, \dots, v_d\}$:

$$\Lambda = \{x_1 v_1 + \dots + x_d v_d : x_i \in \mathbb{Z} \text{ for } i = 1, 2, \dots, d\}$$

Equivalently, $\Lambda = A\mathbb{Z}^d$ for $A = (v_1, v_2, \dots, v_d)$, a nondegenerate $d \times d$ matrix. Packings whose sphere centers form a lattice Λ are called *lattice packings*. Conventionally, for a lattice packing, one uses balls of radius

$$r = \frac{1}{2} \min_{v \in \Lambda \setminus \{0\}} |v|$$

Example 2.1 (2D sphere packing). The lattice packing $\Lambda = \mathbb{Z} + \mathbb{Z}e^{2\pi i/3}$ is optimal in $\mathbb{R}^2$.

A more general notion is that of *periodic packings*, whose centers are of the form $\Lambda + F$, where F is a finite set. Equivalently, these are packings invariant under the action of some lattice Λ and with finitely many orbits. A lattice packing is a periodic packing with one orbit under the action of Λ .

2.1. Density of Lattice Packings. Lattices can be viewed as tiling $\mathbb{R}^d$ by parallelotopes

$$C = \{x_1 v_1 + \cdots + x_d v_d : x_i \in [0, 1) \text{ for } i = 1, 2, \dots, d\}$$

called fundamental cells. In a lattice packing, each cell contains pieces of balls. The density of the packing is

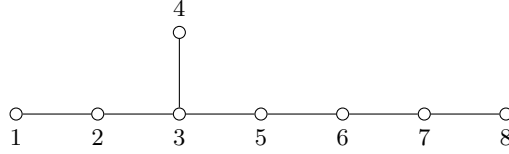
$$\frac{\text{Vol}(B_d(0, r))}{\text{Vol}(C)} = \frac{\pi^{d/2} r^d}{\Gamma(d/2 + 1) \text{Vol}(\mathbb{R}^d/\Lambda)}$$

Corollary 2.2. The density of a periodic packing $\Lambda + F$, where $F = \{t_1, \dots, t_N\}$ and the $t_i + \Lambda$ are disjoint cosets, is

$$\Delta_{\Lambda+F} = \frac{N|B_r|}{|\Lambda|}$$

Proof. The packing $\mathcal{P}_{\Lambda+F}$ is the disjoint union of N packings $\mathcal{P}_{t_i+\Lambda}$. Since density is invariant under translation, each copy has the same density as $\mathcal{P}_\Lambda$. Thus we obtain N copies of the lattice-packing density. $\square$

2.2. E_8 Lattice. We will soon discuss Maryna Viazovska's proof that the optimal packing in dimension 8 is the E_8 lattice packing. The lattice E_8 is defined by the following Dynkin diagram:



This means that E_8 has a basis $\{v_i : 1 \leq i \leq 8\}$ with $\langle v_i, v_i \rangle = 2$, $\langle v_i, v_j \rangle = -1$ if nodes i, j are neighbors, and $\langle v_i, v_j \rangle = 0$ otherwise. Equivalently, its Gram matrix is

$$G = (\langle v_i, v_j \rangle)_{1 \leq i, j \leq 8} = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

One can check by hand that this symmetric matrix is positive definite. Hence, by Cholesky factorization, there exists a lower triangular matrix L such that $G = LL^T$. The rows v_i of L give a basis with the prescribed Gram matrix.

The lattice E_8 is *even*, meaning the squared length of every vector is even:

$$|v|^2 = |m_1 v_1 + m_2 v_2 + \cdots + m_8 v_8|^2 = \sum_i m_i^2 |v_i|^2 + \sum_{i < j} 2m_i m_j \langle v_i, v_j \rangle$$

It is integral, meaning $\langle \cdot, \cdot \rangle$ takes integer values on E_8 . It is also unimodular:

$$\text{Vol}(\mathbb{R}^8/E_8) = 1$$

Question 2.3. Since every $v \in E_8$ has $|v| = \sqrt{2k}$ for some $k \in \mathbb{N}$, does there exist $v \in E_8$ with $|v| = \sqrt{2k}$ for every k ?

Yes. Each basis vector has length $\sqrt{2}$. Squared lengths of orthogonal vectors add, so we want to combine orthogonal vectors. By Lagrange's four-square theorem, every $k \in \mathbb{N}$ can be written as

$$k = a_1^2 + a_2^2 + a_3^2 + a_4^2, \quad a_i \in \mathbb{N}$$

The Dynkin diagram shows that v_1, v_3, v_6, v_8 are mutually orthogonal. Hence

$$v = a_1 v_1 + a_2 v_3 + a_3 v_6 + a_4 v_8$$

has squared length $2k$, and therefore length $\sqrt{2k}$.

Proposition 2.4. E_8 is the unique even unimodular lattice in $\mathbb{R}^8$, up to isometry.

Recall that in linear algebra, for each basis $\{v_1, \dots, v_d\}$ of $\mathbb{R}^d$, there is a unique dual basis $\{v_1^*, \dots, v_d^*\}$ satisfying

$$(3) \quad \langle v_i, v_j^* \rangle = \delta_{ij}$$

Every lattice $\Lambda = \text{sp}_{\mathbb{Z}}(v_1, \dots, v_d)$ has a dual lattice

$$\Lambda^* = \text{sp}_{\mathbb{Z}}(v_1^*, \dots, v_d^*)$$

We can also characterize Λ^* without choosing a basis:

$$(4) \quad \Lambda^* = \{x \in \mathbb{R}^n : \langle x, y \rangle \in \mathbb{Z} \text{ for all } y \in \Lambda\}$$

Indeed, this integrality criterion is satisfied by every element of $\text{sp}_{\mathbb{Z}}(v_1^*, \dots, v_d^*)$ by (3). Conversely, suppose $x \in \mathbb{R}^n$ satisfies this criterion. Write

$$x = a_1 v_1^* + \dots + a_d v_d^*$$

Taking the inner product with v_i gives

$$a_i = \langle x, v_i \rangle \in \mathbb{Z}$$

by (4). Hence

$$x \in \text{sp}_{\mathbb{Z}}(v_1^*, \dots, v_d^*)$$

If $\Lambda = A\mathbb{Z}^d$ with $A = (v_1 | \dots | v_d)$, then

$$\Lambda^* = (A^T)^{-1} \mathbb{Z}^d$$

The covolume of Λ is $\text{Vol}(\mathbb{R}^n/\Lambda)$. Taking determinants on both sides of

$$I = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} - & v_1^* & - \\ - & v_2^* & - \\ \vdots & \vdots & \vdots \\ - & v_n^* & - \end{pmatrix}$$

gives

$$\text{Vol}(\mathbb{R}^n/\Lambda) \text{Vol}(\mathbb{R}^n/\Lambda^*) = 1$$

Proposition 2.5. Every integral unimodular lattice is self-dual: $\Lambda = \Lambda^*$.

Proof. By integrality and (4), we have $\Lambda \subset \Lambda^*$. It remains to show that the index $[\Lambda^* : \Lambda]$ is 1. Since

$$[\Lambda^* : \Lambda] = \frac{\text{Vol}(\mathbb{R}^n/\Lambda)}{\text{Vol}(\mathbb{R}^n/\Lambda^*)} = \text{Vol}(\mathbb{R}^n/\Lambda)^2 = 1$$

we conclude that $\Lambda = \Lambda^*$. □

Packing balls of radius $\sqrt{2}/2$ centered at E_8 gives density

$$\Delta_{E_8} = \frac{\text{Vol}(B_8(0, \sqrt{2}/2))}{\text{Vol}(\mathbb{R}^8/E_8)} = \frac{\pi^4}{384}$$

Viazovska proved in [Via17] that this is optimal. We will discuss the proof later.

2.3. Poisson Summation. For a sufficiently nice function f and a lattice Λ ,

$$(5) \quad \sum_{x \in \Lambda} f(x+t) = \frac{1}{|\Lambda|} \sum_{\xi \in \Lambda^*} \widehat{f}(\xi) e^{2\pi i \langle \xi, t \rangle}$$

Proof. First take $\Lambda = \mathbb{Z}^d$ and $t = 0$. Periodize f by

$$F(x) = \sum_{m \in \mathbb{Z}^d} f(x+m)$$

Then F is $\mathbb{Z}^d$ -periodic. Its Fourier coefficients are

$$(6) \quad \begin{aligned} \widehat{F}(\xi) &= \int_{\mathbb{R}^d/\mathbb{Z}^d} \sum_{m \in \mathbb{Z}^d} f(x+m) e^{-2\pi i \langle \xi, x \rangle} dx \\ &= \sum_{m \in \mathbb{Z}^d} \int_{[0,1)^d+m} f(y) e^{-2\pi i \langle \xi, y \rangle} dy \\ &= \widehat{f}(\xi) \end{aligned}$$

where we used $e^{-2\pi i \langle \xi, m \rangle} = 1$ for $\xi, m \in \mathbb{Z}^d$. Evaluating the Fourier series of F at 0 gives

$$\sum_{x \in \mathbb{Z}^d} f(x) = \sum_{\xi \in \mathbb{Z}^d} \widehat{f}(\xi)$$

Now let $\Lambda = A\mathbb{Z}^d$ and set $g(x) = f(Ax)$. Then

$$\sum_{x \in \mathbb{Z}^d} g(x) = \sum_{y \in \Lambda} f(y), \quad \widehat{g}(\xi) = \frac{1}{|\det A|} \widehat{f}((A^T)^{-1}\xi)$$

Since $\Lambda^* = (A^T)^{-1}\mathbb{Z}^d$ and $|\Lambda| = |\det A|$, applying (7) to g gives

$$\sum_{x \in \Lambda} f(x) = \frac{1}{|\Lambda|} \sum_{\xi \in \Lambda^*} \widehat{f}(\xi)$$

Finally, apply this to $x \mapsto f(x+t)$, whose Fourier transform is $e^{2\pi i \langle \xi, t \rangle} \widehat{f}(\xi)$. This gives (5). $\square$

In the proof, the only analytic point is that we need enough decay to exchange the integral and the infinite sum in (6). For the rest of this article, call f *admissible* if, for some $\epsilon > 0$,

$$\max(|f(x)|, |\widehat{f}(x)|) \leq \frac{C_\epsilon}{(1+|x|)^{d+\epsilon}}$$

This gives absolute convergence of both sums and justifies the exchange by the Weierstrass M -test.

3. LINEAR PROGRAMMING BOUNDS

It seems hopeless to find an upper bound on the density of an arbitrary packing.

Lemma 3.1. Periodic packings come arbitrarily close to optimal packings.

To obtain an upper bound on $\Delta_{\mathbb{R}^d}$, it suffices to upper bound densities of periodic packings. Surprisingly, one way to do this is to find a Schwartz function with special sign properties. The key tool is the Poisson summation formula. The proof of the following theorem is a continuous version of Theorem 1.4.

Theorem 3.2. Suppose there exists $f \in \mathcal{S}(\mathbb{R}^d)$ with the following properties:

- (1) $f(0) > 0$ and $\hat{f}(\xi) \geq 0$ for all $\xi \in \mathbb{R}^d$
- (2) $f(x) \leq 0$ for $|x| \geq r$

Then

$$\Delta_d \leq \frac{f(0)}{\hat{f}(0)} \text{Vol}(B_{r/2})$$

Proof. We remark that $f(0) > 0$ is implied by $\hat{f}(\xi) \geq 0$ and $\hat{f} \not\equiv 0$.

Step 1: Lattice packings. Let Λ be a lattice, and let f have the above properties. By rescaling, which does not change the density, we may assume the shortest nonzero vector in Λ has length r . The lattice packing with balls of radius $r/2$ has density

$$\Delta_\Lambda = \frac{|B_{r/2}|}{|\Lambda|}$$

The desired conclusion is equivalent to showing

$$\frac{f(0)}{\hat{f}(0)} |\Lambda| \geq 1$$

By Poisson summation,

$$\sum_{x \in \Lambda} f(x) = \frac{1}{|\Lambda|} \sum_{\xi \in \Lambda^*} \hat{f}(\xi)$$

On the one hand, by our rescaling assumption,

$$\text{LHS} \leq f(0)$$

On the other hand, since $\hat{f} \geq 0$, we can drop every term on the right-hand side except $\hat{f}(0)$:

$$\text{RHS} \geq \frac{\hat{f}(0)}{|\Lambda|}$$

Combining the two inequalities gives the desired bound.

Step 2: Periodic packings. A periodic packing with balls of radius R centered at $\Lambda + F$, where $F = \{t_1, \dots, t_N\}$, has density

$$\Delta_{\Lambda+F} = N \frac{|B_R|}{|\Lambda|}$$

By rescaling, we may again assume that the shortest nonzero vector in $\Lambda + F$ has length r . Hence the desired claim is equivalent to

$$\frac{f(0)}{\hat{f}(0)} |\Lambda| \geq N$$

We now sum the translated Poisson formula (5) and use the same sign argument:

$$\begin{aligned}
 Nf(0) &\geq \sum_{i,j=1}^N \sum_{x \in \Lambda} f(x + t_i - t_j) \\
 &= \frac{1}{|\Lambda|} \sum_{i,j=1}^N \sum_{\xi \in \Lambda^*} \hat{f}(\xi) e^{2\pi i \langle \xi, t_i \rangle} e^{-2\pi i \langle \xi, t_j \rangle} \\
 &= \frac{1}{|\Lambda|} \sum_{\xi \in \Lambda^*} \hat{f}(\xi) \left| \sum_{i=1}^N e^{2\pi i \langle \xi, t_i \rangle} \right|^2 \\
 &\geq \frac{N^2 \hat{f}(0)}{|\Lambda|}
 \end{aligned}$$

This gives the desired inequality. $\square$

3.1. One-Dimensional Optimal Packing. As a toy example, we use Theorem 3.2 in dimension 1. Consider

$$f(x) = \chi_{[-1/2, 1/2]} * \chi_{[-1/2, 1/2]} = (1 - |x|)\chi_{[-1, 1]}$$

By the convolution theorem,

$$\hat{f}(\xi) = (\hat{\chi}_{[-1/2, 1/2]})^2 = \left(\frac{\sin \pi \xi}{\pi \xi} \right)^2$$

We have $f(0) = \hat{f}(0) = 1 > 0$, $\hat{f} \geq 0$, and $f(x) \leq 0$ for $|x| \geq 1$. Hence the hypotheses of Theorem 3.2 are satisfied with $r = 1$. Therefore the optimal packing density in dimension 1 is upper bounded by

$$\text{Vol}(B_1(0, 1/2)) = 1$$

3.2. Magic Functions. Can the bound for lattice packings in Theorem 3.2 be sharp for some $f \in \mathcal{S}$? Recall that for f obeying the conditions of Theorem 3.2,

$$f(0) \geq \sum_{x \in \Lambda} f(x) = \frac{1}{|\Lambda|} \sum_{\xi \in \Lambda^*} \hat{f}(\xi) \geq \frac{\hat{f}(0)}{|\Lambda|}$$

Equality is attained in the first and last inequalities if and only if $f = 0$ on $\Lambda \setminus \{0\}$ and $\hat{f} = 0$ on $\Lambda^* \setminus \{0\}$. Such functions are called *magic functions*.

Does a magic function f exist? In other words, are linear programming bounds sharp? The answer is: not always. However, numerical evidence suggests that in dimensions 8 and 24,

Theorem 3.3 (Cohn–Elkies 2003).

$$\Delta_8 \leq 1.00016 \Delta_{E_8}, \quad \Delta_{24} \leq 1.019 \Delta_{\Lambda_{24}}$$

The known upper bounds given by 3.2 and the known lower bounds are extremely close (Fig 1). It is tempting to conjecture that a magic function occurs in these dimensions.

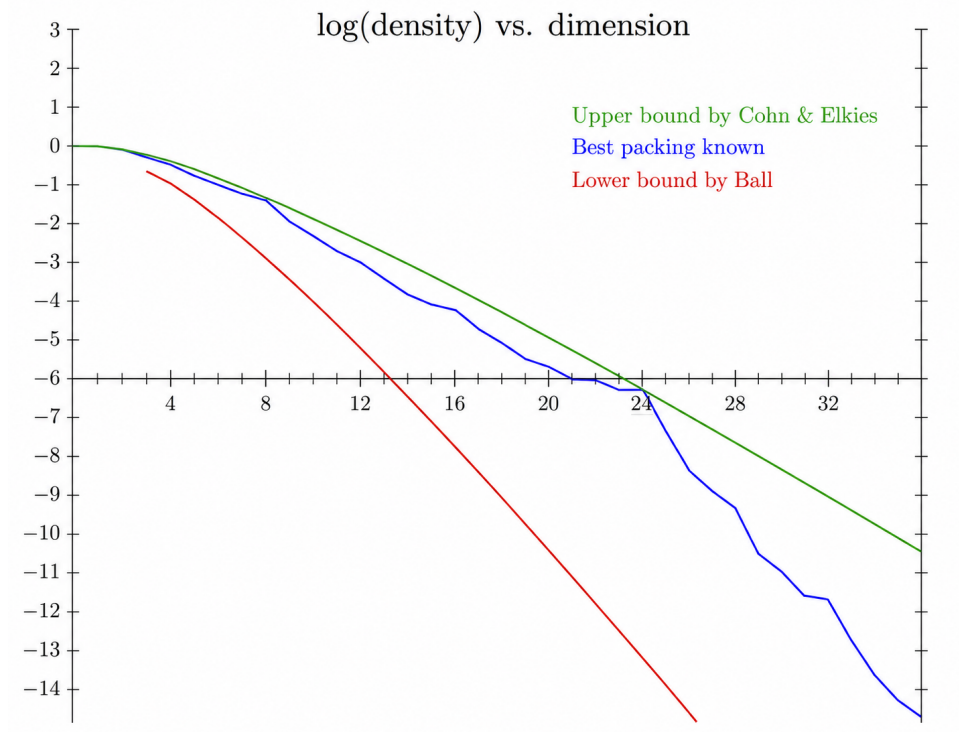


FIGURE 1. Numerical Bounds

4. MAGIC FUNCTIONS

The rotational symmetry of the requirements in Theorem 3.2 suggests that we should search for a radial magic function. We desire to find a radial Schwartz function $g : \mathbb{R}^8 \rightarrow \mathbb{R}$ satisfying

$$(7) \quad g(0) = \hat{g}(0) = 1, \quad g(x) \leq 0 \text{ for } \|x\| \geq \sqrt{2}, \quad \hat{g}(\xi) \geq 0 \text{ for all } \xi \in \mathbb{R}^8$$

Moreover, sharpness for the E_8 lattice forces both g and $\hat{g}$ to vanish on

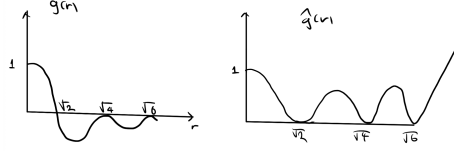
$$E_8 \setminus \{0\}$$

Since the squared lengths of vectors in E_8 lie in $2\mathbb{Z}_{\geq 0}$, the desired zeros occur at radii

$$r = \sqrt{2k}, \quad k \in \mathbb{N}$$

For $r > \sqrt{2}$, these zeros should be double zeros: they occur inside the region where $g \leq 0$ or $\hat{g} \geq 0$, so the derivative must vanish as well. The first zero $r = \sqrt{2}$ is exceptional, since this is where g changes sign. See Figure 2.

In this section, we explain the Fourier-analytic viewpoint behind Viazovska's ansatz for g as a Laplace-type transform. The guiding idea is to build g and $\hat{g}$ with parallel structures, so that their zeros can be controlled simultaneously. This requirement leads naturally to functional equations involving the modular transformations $z \mapsto z+1$ and $z \mapsto -1/z$. Finally, we briefly sketch how Viazovska engineers the required modular forms to obtain the explicit magic function.


 FIGURE 2. A schematic sketch of g and $\widehat{g}$.

4.1. Fourier eigenfunctions. Our first goal is to control the zeros of g and $\widehat{g}$ simultaneously. A useful observation is that if g is a radial magic function satisfying (7), then

$$f_1(x) := \frac{\widehat{g}(x) + g(x)}{2}, \quad f_2(x) := \frac{\widehat{g}(x) - g(x)}{2}$$

are Fourier eigenfunctions. Indeed, since g is radial, $\widehat{\widehat{g}} = g$. Hence

$$\widehat{f_1} = f_1, \quad \widehat{f_2} = -f_2$$

Conversely, we recover g and $\widehat{g}$ by

$$g = f_1 - f_2, \quad \widehat{g} = f_1 + f_2$$

Thus, rather than constructing g directly, it is natural to search for two Fourier eigenfunctions a and b satisfying

$$\widehat{a} = a, \quad \widehat{b} = -b$$

Then a linear combination

$$g(x) := c_1 a(x) + c_2 b(x)$$

has Fourier transform

$$\widehat{g}(x) = c_1 a(x) - c_2 b(x)$$

The constants c_1, c_2 can then be chosen to normalize $g(0)$ and $\widehat{g}(0)$, while the construction of a and b is designed to impose the required zeros (with multiplicity) for $g(x)$ and $\widehat{g}(x)$.

Reduction: search for Fourier eigenfunctions a, b with $\widehat{a} = a$, $\widehat{b} = -b$.

The reason modular forms enter the proof is already visible from the Fourier transform of a Gaussian. In dimension 8, for $t > 0$,

$$\mathcal{F}\left(e^{-\pi t \|x\|^2}\right)(\xi) = t^{-4} e^{-\pi \|\xi\|^2 / t}$$

This suggests looking for Fourier eigenfunctions built as continuous linear combinations of Gaussians. Suppose, formally, that

$$b(x) = \int_0^\infty \varphi(t) e^{-\pi t \|x\|^2} dt$$

Assuming that φ has sufficient decay to justify interchanging the Fourier transform and the integral, we obtain

$$\widehat{b}(\xi) = \int_0^\infty \varphi(t) t^{-4} e^{-\pi \|\xi\|^2 / t} dt$$

After the change of variables $t \mapsto 1/t$, this becomes

$$\widehat{b}(\xi) = \int_0^\infty t^2 \varphi(1/t) e^{-\pi t \|\xi\|^2} dt$$

Thus $\widehat{b} = \pm b$ would follow if

$$\varphi(t) = \pm t^2 \varphi(1/t)$$

This self-similarity under $t \mapsto 1/t$ is the first hint that modular forms should appear. To make the relation with modular transformations more explicit, write $\psi(it) = \varphi(t)$ and set $z = it$. Then the condition above becomes,

$$\begin{aligned} a = \pm \widehat{a} &\iff \int_0^\infty \psi(it) e^{-\pi t \|x\|^2} dt = \pm \int_0^\infty t^2 \psi(i/t) e^{-\pi t \|x\|^2} dt \\ &\iff \int_0^{i\infty} \psi(z) e^{\pi i \|x\|^2 z} dz = \mp \int_0^{i\infty} z^2 \psi\left(-\frac{1}{z}\right) e^{\pi i \|x\|^2 z} dz \end{aligned}$$

Hence the condition for a +1 Fourier eigenfunction has the schematic form

$$(8) \quad \psi(z) = -z^2 \psi\left(-\frac{1}{z}\right)$$

whereas the condition for a -1 Fourier eigenfunction has the schematic form

$$(9) \quad \psi(z) = z^2 \psi\left(-\frac{1}{z}\right)$$

These are *modular* transformation laws. In particular, (9) is the transformation law of a weight -2 modular form under the map $S : z \mapsto -1/z$, we will explain relevant definitions now.

4.2. A Crash Course on Modular Forms. Only a small part of the theory of modular forms is needed for our purposes. The main point is that modular forms are holomorphic functions on the upper half-plane

$$\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$$

with controlled behavior under the two transformations

$$S : z \mapsto -\frac{1}{z}, \quad T : z \mapsto z + 1$$

These are exactly the transformations that appeared in the Gaussian heuristic: the Fourier transform of a Gaussian sends the parameter by S , while the sine-square factor later produces shifts by T .

A motivating example is the *theta series* of a lattice Λ :

$$\theta_\Lambda(z) = \sum_{x \in \Lambda} e^{\pi i \|x\|^2 z}$$

This is a generating function for the squared lengths of vectors in Λ . Applying Poisson summation to the Gaussian

$$f(x) = e^{\pi i \|x\|^2 z}$$

gives

$$\theta_\Lambda(z) = \frac{1}{\text{vol}(\mathbb{R}^n/\Lambda)} \left(\frac{i}{z}\right)^{n/2} \theta_{\Lambda^*}\left(-\frac{1}{z}\right)$$

For E_8 , we have $E_8^* = E_8$ and $\text{vol}(\mathbb{R}^8/E_8) = 1$, so

$$\theta_{E_8}\left(-\frac{1}{z}\right) = z^4 \theta_{E_8}(z)$$

Moreover, since E_8 is even, $\|x\|^2 \in 2\mathbb{Z}$, and hence

$$\theta_{E_8}(z+1) = \theta_{E_8}(z)$$

This example already shows the two basic modular symmetries.

Let

$$\Gamma(1) = PSL_2(\mathbb{Z})$$

be the modular group. It acts on $\mathbb{H}$ by linear fractional transformations

$$\gamma z = \frac{az+b}{cz+d}, \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

For a function F on $\mathbb{H}$, the weight k slash operator is

$$(F|_k \gamma)(z) := (cz+d)^{-k} F(\gamma z)$$

This notation records the automorphy factor appearing under a change of coordinates.

Definition 4.1 (Modular form). Let $\Gamma \subseteq \Gamma(1)$ be a finite-index subgroup. A holomorphic modular form of weight k for Γ is a holomorphic function

$$f : \mathbb{H} \rightarrow \mathbb{C}$$

such that

$$f|_k \gamma = f \quad \text{for all } \gamma \in \Gamma,$$

and such that f is holomorphic at the cusps. Equivalently, for every $\alpha \in \Gamma(1)$, the transformed function $f|_k \alpha$ has a Fourier expansion

$$(f|_k \alpha)(z) = \sum_{n=0}^{\infty} c_{\alpha}(n) e^{2\pi i n z / h_{\alpha}}$$

for some positive integer h_{α} , called the width of the cusp.

The cusps are rational boundary points $\mathbb{Q} \cup \{\infty\}$, modulo the action of Γ . To check the behavior at a cusp, one moves it to ∞ by some $\alpha \in \Gamma(1)$ and then studies $f|_k \alpha$ as $\text{Im } z \rightarrow \infty$. The possible non-integer frequencies n/h_{α} simply mean that the period at that cusp may be h_{α} , rather than 1. We write $M_k(\Gamma)$ for the space of modular forms of weight k for Γ .

A *weakly holomorphic modular form* satisfies the same modular transformation law, but its Fourier expansion at a cusp may have finitely many negative powers:

$$(f|_k \alpha)(z) = \sum_{n=n_0}^{\infty} c_{\alpha}(n) e^{2\pi i n z / h_{\alpha}}, \quad n_0 \in \mathbb{Z}$$

Thus weakly holomorphic means that poles at cusps are allowed. This is essential below because holomorphic modular forms of negative weight vanish, whereas Viazovska needs nontrivial forms of weights -2 and -4 .

A *quasimodular form* is different: it is holomorphic on $\mathbb{H}$, but its modular transformation law has controlled error terms. The key example is E_2 , introduced below. This distinction will matter later: weakly holomorphic refers to cusp poles, while quasimodular refers to controlled failure of the modular transformation law.

Example 4.2 (Eisenstein series). For even $k \geq 4$, the Eisenstein series is

$$E_k(z) := \frac{1}{2\zeta(k)} \sum_{(c,d) \in \mathbb{Z}^2 \setminus \{(0,0)\}} (cz + d)^{-k}$$

It is a modular form of weight k for $\Gamma(1)$. With $q = e^{2\pi iz}$, it has Fourier expansion

$$E_k(z) = 1 + \frac{2}{\zeta(1-k)} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n, \quad \sigma_{k-1}(n) = \sum_{d|n} d^{k-1}$$

In particular,

$$E_4(z) = 1 + 240 \sum_{n=1}^{\infty} \sigma_3(n) q^n, \quad E_6(z) = 1 - 504 \sum_{n=1}^{\infty} \sigma_5(n) q^n$$

The analogous series for $k = 2$ is not modular. We define

$$E_2(z) = 1 - 24 \sum_{n=1}^{\infty} \sigma_1(n) q^n,$$

and it satisfies the quasimodular transformation law

$$z^{-2} E_2\left(-\frac{1}{z}\right) = E_2(z) - \frac{6i}{\pi z}$$

This explicit error term is precisely why E_2 is useful in Viazovska's construction.

Example 4.3 (The j -invariant). The function

$$j(z) = \frac{1728 E_4(z)^3}{E_4(z)^3 - E_6(z)^2}$$

is a weakly holomorphic modular form of weight 0. Its Fourier expansion begins

$$j(z) = q^{-1} + 744 + 196884q + \dots$$

The negative power q^{-1} means that j has a pole at the cusp.

Example 4.4 (Theta functions). Viazovska also uses the theta functions

$$\begin{aligned} \theta_{00}(z) &= \sum_{n \in \mathbb{Z}} e^{\pi i n^2 z}, & \theta_{01}(z) &= \sum_{n \in \mathbb{Z}} (-1)^n e^{\pi i n^2 z}, \\ \theta_{10}(z) &= \sum_{n \in \mathbb{Z}} e^{\pi i (n+1/2)^2 z} \end{aligned}$$

Their fourth powers have especially simple transformation laws under S and T :

$$z^{-2} \theta_{00}^4\left(-\frac{1}{z}\right) = -\theta_{00}^4(z), \quad z^{-2} \theta_{01}^4\left(-\frac{1}{z}\right) = -\theta_{10}^4(z), \quad z^{-2} \theta_{10}^4\left(-\frac{1}{z}\right) = -\theta_{01}^4(z),$$

and

$$\theta_{00}^4(z+1) = \theta_{01}^4(z), \quad \theta_{01}^4(z+1) = \theta_{00}^4(z), \quad \theta_{10}^4(z+1) = -\theta_{10}^4(z)$$

Thus the theta functions form a small system that is permuted by the same transformations appearing in the contour calculation.

4.3. The Magic Integral Transform. The previous discussion does not yet impose the desired zeros. Viazovska's bold idea is to insert these roots directly by multiplying the Gaussian integral by a sine-squared prefactor. Formally, consider an ansatz of the form

$$(10) \quad f_\varphi(r) = -4 \sin^2 \left(\frac{\pi r^2}{2} \right) \int_0^{i\infty} \varphi(z) e^{\pi i r^2 z} dz$$

The factor

$$\sin^2 \left(\frac{\pi r^2}{2} \right)$$

has double zeros at

$$r = \sqrt{2k}, \quad k \in \mathbb{N}$$

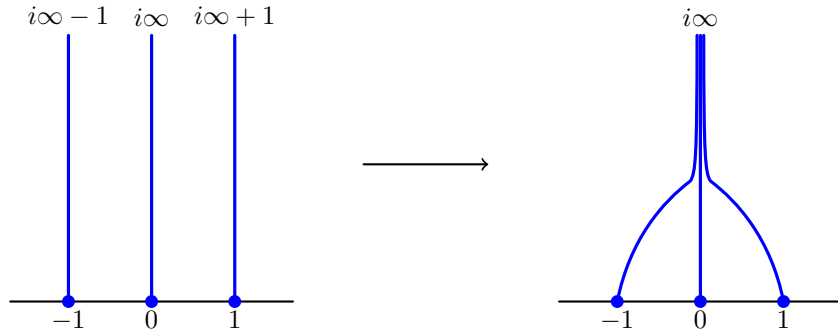
which matches the desired zero multiplicity at the E_8 -lattice radii, except at $r = 0$ and at the minimal radius $r = \sqrt{2}$.

Expanding sin in terms of complex exponentials, we can rewrite (10) as

$$\begin{aligned} f_\varphi(r) &= \int_0^{i\infty} \varphi(z) \left(e^{\pi i r^2 (z+1)} - 2e^{\pi i r^2 z} + e^{\pi i r^2 (z-1)} \right) dz \\ &= \int_1^{1+i\infty} \varphi(z-1) e^{\pi i r^2 z} dz - 2 \int_0^{i\infty} \varphi(z) e^{\pi i r^2 z} dz \\ &\quad + \int_{-1}^{-1+i\infty} \varphi(z+1) e^{\pi i r^2 z} dz \end{aligned}$$

Thus the sine-squared factor converts a single vertical contour into three shifted vertical contours.

Ignoring convergence issues for the moment, let us deform the three contours to a “chicken claw” contour.



Define the second finite difference

$$\tilde{\varphi}(z) = \varphi(z+1) - 2\varphi(z) + \varphi(z-1)$$

Then the deformed expression has the schematic form

$$(11) \quad \begin{aligned} f_\varphi(r) &= \int_{-1}^i \varphi(z+1) e^{\pi i r^2 z} dz - 2 \int_0^i \varphi(z) e^{\pi i r^2 z} dz + \int_1^i \varphi(z-1) e^{\pi i r^2 z} dz \\ &\quad + \int_i^{i\infty} \tilde{\varphi}(z) e^{\pi i r^2 z} dz \end{aligned}$$

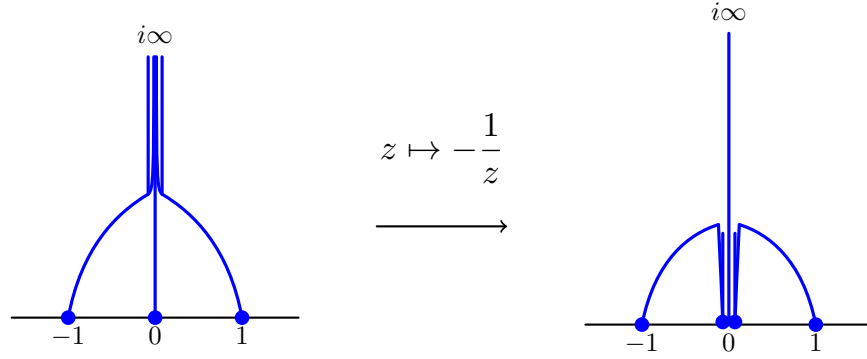
We now compute the Fourier transform formally, assuming that the Fourier transform may be interchanged with the contour integrals. In dimension 8,

$$\mathcal{F}\left(e^{\pi i z \|x\|^2}\right)(\xi) = z^{-4} e^{\pi i(-1/z)\|\xi\|^2}$$

Applying this to (11) gives the same contour expression, but with the Gaussian parameter changed by

$$z \mapsto -\frac{1}{z}$$

After making the change of variables $z \mapsto -1/z$, one obtains the “inverse chicken claw contour”



$$\begin{aligned} \widehat{f}_\varphi(r) &= \int_1^i \varphi\left(-\frac{1}{z} + 1\right) z^2 e^{\pi i r^2 z} dz - 2 \int_{i\infty}^i \varphi\left(-\frac{1}{z}\right) z^2 e^{\pi i r^2 z} dz \\ (12) \quad &+ \int_{-1}^i \varphi\left(-\frac{1}{z} - 1\right) z^2 e^{\pi i r^2 z} dz + \int_i^0 \widetilde{\varphi}\left(-\frac{1}{z}\right) z^2 e^{\pi i r^2 z} dz \end{aligned}$$

We would like $\widehat{f}_\Lambda$ to have the same form as (10). Therefore we compare (12) with

$$f_\psi(r) = -4 \sin^2\left(\frac{\pi r^2}{2}\right) \int_0^{i\infty} \psi(z) e^{\pi i r^2 z} dz$$

This leads formally to the matching conditions

$$(13) \quad \psi(z+1) = \varphi\left(-\frac{1}{z} - 1\right) z^2,$$

$$(14) \quad \psi(z-1) = \varphi\left(-\frac{1}{z} + 1\right) z^2,$$

$$(15) \quad \widetilde{\psi}(z) = 2\varphi\left(-\frac{1}{z}\right) z^2,$$

$$(16) \quad 2\psi(z) = \widetilde{\varphi}\left(-\frac{1}{z}\right) z^2.$$

If these identities hold, then the Fourier transform of f_φ has the same form as f_ψ :

$$\widehat{f}_\varphi = f_\psi$$

Thus, to obtain a $+1$ eigenfunction, one would further need $\psi = \varphi$, while to obtain a -1 eigenfunction, one would need $\psi = -\varphi$.

Viazovska's Section 4 carries out this strategy by constructing two radial Schwartz functions

$$a, b : \mathbb{R}^8 \rightarrow i\mathbb{R}$$

with

$$\widehat{a} = a, \quad \widehat{b} = -b$$

For $r > \sqrt{2}$, these functions admit the representations

$$a(r) = -4 \sin^2 \left(\frac{\pi r^2}{2} \right) \int_0^{i\infty} \phi_0 \left(-\frac{1}{z} \right) z^2 e^{\pi i r^2 z} dz,$$

and

$$b(r) = -4 \sin^2 \left(\frac{\pi r^2}{2} \right) \int_0^{i\infty} \psi_I(z) e^{\pi i r^2 z} dz$$

Here ϕ_0 is built from Eisenstein series, E_2 , and the j -invariant, while ψ_I is built from theta functions. These formulas exhibit the prescribed double zeros at all radii $r = \sqrt{2k}$ with $k \geq 2$.

The final magic function is constructed as the linear combination

$$(17) \quad g(x) = \frac{\pi i}{8640} a(x) + \frac{i}{240\pi} b(x)$$

Remark 4.5. The identities (13)–(16) should be read as a heuristic guide, not as the literal system imposed in Viazovska's paper. It is true that the integrands should have controlled behavior under $S : z \mapsto -1/z$ and $T : z \mapsto z + 1$. But the actual construction is more delicate: the chosen forms must give the correct zero multiplicities at the E_8 -radii at 0 and $\sqrt{2}$ needed for $g(x)$ and $\widehat{g}(x)$.

To this end, what we will need are modular forms with "poles" (weakly holomorphic) or have controlled error terms in modular transformation laws (quasi-modular).

4.4. An Overview of Viazovska's Construction. We now summarize the structural wishlist achieved by Viazovska's construction. Since the full calculation is delicate, we focus on the main characters involved and the motivation for their choice.

(1) Eigenfunctions:

$$\widehat{a}(x) = a(x), \quad \widehat{b}(x) = -b(x)$$

(2) Zeros: a and b both will have double zeros at $\sqrt{2n}$ for $n \geq 2$. Moreover,

$$a(0) \sim 1, \quad a(\sqrt{2}) = 0, \quad a'(\sqrt{2}) \neq 0,$$

and

$$b(0) = 0, \quad b(\sqrt{2}) = 0, \quad b'(\sqrt{2}) \neq 0$$

The coefficients in the final linear combination are chosen so that

$$g'(\sqrt{2}) \neq 0, \quad \widehat{g}'(\sqrt{2}) = 0$$

Thus g has a simple zero at the first shell, allowing the required sign change, while $\widehat{g}$ has a double zero there.

(3) Signs:

$$g(x) = c_1 a(x) + c_2 b(x) \leq 0 \quad \text{for } \|x\| \geq \sqrt{2},$$

and

$$\widehat{g}(x) = c_1 a(x) - c_2 b(x) \geq 0 \quad \text{for all } x \in \mathbb{R}^8$$

(4) Normalization:

$$g(0) = \widehat{g}(0) = 1$$

The construction breaks into two packages: a uses E_2, E_4, E_6, j from Examples 4.2 and 4.3, while b uses theta functions from Example 4.4. The special values of a and b allow the constants c_1, c_2 to be chosen so that $g(0) = \widehat{g}(0) = 1$ and the first-shell derivatives have the required behavior.

Construction of b . For the b -function, Viazovska uses theta functions. Recall from the comment below (9) that to construct -1 eigenfunctions, we need a weakly holomorphic modular form of weight -2 .¹

As in Example 4.4, the fourth powers

$$\theta_{00}^4, \quad \theta_{01}^4, \quad \theta_{10}^4$$

are explicitly permuted by the modular transformations

$$S : z \mapsto -\frac{1}{z}, \quad T : z \mapsto z + 1$$

This makes them well suited to the contour calculation, where different branches must transform into one another. One can construct a weight -2 weakly holomorphic modular form with the desired pole at the cusp by considering

$$h = 128 \frac{\theta_{00}^4 + \theta_{01}^4}{\theta_{10}^8}$$

From h , Viazovska defines

$$\psi_I := h - h|_{-2}ST$$

The -1 -eigenfunction b has the sine-square representation

$$b(r) = -4 \sin^2 \left(\frac{\pi r^2}{2} \right) \int_0^{i\infty} \psi_I(z) e^{\pi i r^2 z} dz$$

The pole at the cusp controls the first-shell multiplicity. The q^{-1} -term in ψ_I gives

$$\psi_I(it) \sim e^{2\pi t} \quad (t \rightarrow \infty),$$

so the tail of the integral behaves like

$$\int_0^\infty e^{-\pi(r^2-2)t} dt \sim \frac{1}{\pi(r^2-2)}$$

Thus the integral has a simple pole at $r^2 = 2$. Multiplying by

$$\sin^2 \left(\frac{\pi r^2}{2} \right) \sim \frac{\pi^2}{4} (r^2 - 2)^2$$

cancels one order of this double zero, leaving a simple zero at $r = \sqrt{2}$. This is why the cusp pole is useful: it produces the exceptional first-shell behavior, while the sine-square factor supplies the prescribed double zeros at $r = \sqrt{2n}$, $n \geq 2$.

¹Weakly holomorphic is needed because ordinary holomorphic modular forms of negative weight vanish; allowing poles at the cusps gives nontrivial negative-weight forms.

Construction of a . The a -function is more subtle. The Gaussian heuristic earlier suggests that a +1 Fourier eigenfunction should obey a transformation law of the form

$$\psi(z) = -z^2 \psi\left(-\frac{1}{z}\right)$$

This is modular in spirit, but it is not supplied by the theta-function package used for b . Motivationally, Viazovska starts from the zero-revealing ansatz

$$d(r) := -4 \sin^2\left(\frac{\pi r^2}{2}\right) \int_0^{i\infty} \phi_0\left(-\frac{1}{z}\right) z^2 e^{\pi i r^2 z} dz$$

Formally expanding the sine-square factor and deforming contours as before gives

$$\begin{aligned} d(r) = & \int_{-1}^i \phi_0\left(-\frac{1}{z+1}\right) (z+1)^2 e^{\pi i r^2 z} dz - 2 \int_0^i \phi_0\left(-\frac{1}{z}\right) z^2 e^{\pi i r^2 z} dz \\ & + \int_1^i \phi_0\left(-\frac{1}{z-1}\right) (z-1)^2 e^{\pi i r^2 z} dz \\ & + \int_i^{i\infty} \left[\phi_0\left(-\frac{1}{z+1}\right) (z+1)^2 - 2\phi_0\left(-\frac{1}{z}\right) z^2 + \phi_0\left(-\frac{1}{z-1}\right) (z-1)^2 \right] e^{\pi i r^2 z} dz. \end{aligned}$$

The matching condition (16) motivates setting the final bracket equal to $2\phi_0(z)$. Equivalently, if

$$\Psi(z) := z^2 \phi_0\left(-\frac{1}{z}\right),$$

we want

$$\Psi(z+1) - 2\Psi(z) + \Psi(z-1) = 2\phi_0(z)$$

This is where quasimodularity is used.² Viazovska introduces a weakly holomorphic quasimodular package

$$\phi_{-4}, \quad \phi_{-2}, \quad \phi_0,$$

where the subscripts indicate weights. Concretely,

$$\phi_{-4} := \varphi_{-4}, \quad \phi_{-2} := \varphi_{-4} E_2 + \varphi_{-2},$$

and

$$\phi_0 = \varphi_{-4} E_2^2 + 2\varphi_{-2} E_2 + j - 1728$$

Here j is the weakly holomorphic weight 0 form from Example 4.3. As in Example 4.2, the role of E_2 is that its failure to be modular is explicitly controlled:

$$z^{-2} E_2 \left(-\frac{1}{z}\right) = E_2(z) - \frac{6i}{\pi z}$$

This lets Viazovska engineer the transformation law

$$\phi_0\left(-\frac{1}{z}\right) = \phi_0(z) - \frac{12i}{\pi z} \phi_{-2}(z) - \frac{36}{\pi^2 z^2} \phi_{-4}(z)$$

Hence

$$\Psi(z) = z^2 \phi_0(z) - \frac{12i}{\pi} z \phi_{-2}(z) - \frac{36}{\pi^2} \phi_{-4}(z)$$

²An ordinary holomorphic modular form of weight 0 would be constant, while a weakly holomorphic weight 0 modular form would not have the desired cusp behavior. Viazovska instead uses a weight 0-type quasimodular form.

Since $\phi_0, \phi_{-2}, \phi_{-4}$ are 1-periodic, the second difference gives

$$(z+1)^2 - 2z^2 + (z-1)^2 = 2,$$

while the lower-degree correction terms vanish:

$$(z+1) - 2z + (z-1) = 0, \quad 1 - 2 + 1 = 0$$

Therefore

$$\Psi(z+1) - 2\Psi(z) + \Psi(z-1) = 2\phi_0(z)$$

Thus the quasimodular error terms are useful precisely because, after multiplication by z^2 , they become linear and constant terms killed by the sine-square second difference. The term $j - 1728$ tunes the q -expansion so that ϕ_0 has the desired decay at the cusp.

4.5. Conclusion. Using the functions a and b constructed above, Viazovska proves that the linear combination

$$g(x) = \frac{\pi i}{8640} a(x) + \frac{i}{240\pi} b(x)$$

is a radial Schwartz magic function satisfying the requirements in (7). The sine-square representations explain the prescribed zeros at the E_8 -radii, while the remaining sign conditions are verified by reducing them to explicit one-variable inequalities. Applying the Cohn–Elkies theorem then gives

$$\Delta_{\mathbb{R}^8} = \Delta_{E_8} = \frac{\pi^4}{384}.$$

This proves the optimality of the E_8 lattice packing in dimension 8.

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